

1. (a) Solve the system: 
$$\begin{cases} x + y = 5 \\ x - y = 4 \end{cases}$$
  
  
(b) Solve for  $x$  and  $y$ :  $2x + 3y = 7$ ,  
 $4x - y = 14$
2. The inverse of an invertible matrix never exists.  
(A) True  
(B) False
3. The determinant of a diagonal matrix equals the sum of its diagonal entries. from Q5  
(A) True  
(B) False
4. Answer True or False for each of the following statements:
  - (a) The determinant of a triangular matrix is equal to the product of its diagonal entries.  
(A) True  
(B) False
  - (b) If a matrix  $A$  is invertible, then its determinant must be zero.  
(A) True  
(B) False
  - (c) The trace of a matrix is always equal to the sum of its eigenvalues.  
(A) True  
(B) False
  - (d) The transpose of a symmetric matrix is equal to itself.  
(A) True  
(B) False